

Fermionic Extensions of Continuous Functional Tensor Networks: Exchange-Rank and Exterior-Contraction Obstructions with Restricted Numerical Consequences

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Abstract

Hong et al. proposed continuous functional tensor networks by expanding each particle coordinate in an orthonormal functional basis, representing the coefficient tensor by a tensor network, evaluating basis-projected operators by contraction, and optimizing the tensors by automatic differentiation. We study the fermionic extension that their work left open. Two independent obstructions emerge. First, an ordinary particle-site tensor train pays an unavoidable binomial bond for exchange statistics; we recall the known flat particle spectrum of a single Slater determinant. Second, moving antisymmetry into exterior algebra removes that representation floor but does not make compact inputs easy to contract: exact squared-norm evaluation is $\#P$ -hard already at fixed one-form bond two under a polynomial-time metric reduction, and an independent reduction gives exact point-evaluation hardness in an explicit rational shifted-Legendre functional basis at the same bond. Along a matrix-pair subfamily, bounded and selected fixed-state memories collapse to polynomial linear combinations of established antisymmetrized geminal powers, while a bandwidth-one unique path already encodes a permanent. A generic relative squared-norm scheme would also approximate real positive-semidefinite permanents, whereas entrywise-nonnegative and additive promised cases remain open. Finally, cut-rank divisibility rejects the proposed universal direct tensor product of a statistics carrier and a Slater-normalized correlation multiplicity. Exact certificates support the proof audit. A nonbranching exterior subclass is also evaluated as a numerical consequence of these constraints. Because that subclass is algebraically a finite nonorthogonal Slater expansion, the data are not claimed as a method distinct from nonorthogonal configuration interaction (NOCI). These exact obstructions constrain, but do not exclude, genuinely distinct restricted or controlled approximate algorithms; they do not establish a scalable generic fermionic solver.

1 Scope and thesis

The direct motivation is the functional-tensor-network construction of Hong–Xiao–Hu–Ji–Ran [1]. For orthonormal local functions $\{\phi_s\}_{s=1}^D$, a first-quantized wavefunction is written

$$\Psi(x_1, \dots, x_N) = \sum_{s_1, \dots, s_N} C_{s_1 \dots s_N} \prod_{i=1}^N \phi_{s_i}(x_i). \quad (1)$$

Orthonormality turns the continuum norm into the Euclidean norm of C and turns differential and multiplicative operators into finite functional-basis matrices. Hong et al. compressed C by an MPS and optimized its Rayleigh quotient with automatic differentiation; they explicitly named fermionic tensor networks as a future electronic extension.

For identical fermions, however, $C \in \wedge^N V_D$. An ordinary MPS over the labelled particle slots and an occupation-number MPS use different tensor product factorizations. Their bond dimensions are not related by the abstract Hilbert-space isomorphism. Hilbert-space MPS [2], Grassmann/graded networks [3], and the ordered first-quantized construction of Li–Waintal [4] are therefore comparators, not aliases for the representation studied here.

Throughout, a *theorem* is a formal statement proved in the manuscript. Exact-arithmetic checks are reported only as bounded reproducibility checks and never replace a proof. Floating-point calculations are explicitly described as numerical evidence, with basis, truncation, and optimization controls. Unproved algorithmic consequences are called conjectures. Exact anti-symmetry is structural in every exterior or ordered-chamber construction below; its reported residual is $\epsilon_{\text{AS}} = 0$. The generic exact-contraction obstruction motivates restricted and controlled approximate algorithms rather than excluding them.

The thesis is deliberately negative and representation-specific:

Ordinary particle TT stores exchange multiplicity, whereas compact exterior cores can hide hard coefficient evaluation. Removing the former cost does not remove the latter, and neither observation is a no-go theorem for every fermionic tensor network, ordered sector, determinant expansion, or promised approximation.

2 Exchange rank in the labelled-particle factorization

Let V be a finite-dimensional real or complex Hilbert space. Identify $\Lambda^N V$ isometrically with the alternating subspace of $V^{\otimes N}$ by

$$J_N(v_1 \wedge \cdots \wedge v_N) = \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(N)}.$$

Write $C_{(k)}$ for the $k \mid N - k$ particle matricization.

Theorem 2.1 (Structural result I: exact particle-TT ranks). *For $1 \leq k < N$, the minimal exact ordinary-TT bond is*

$$r_k^{\text{TT}}(C) = \text{rank } C_{(k)}.$$

Proof. Write an arbitrary TT coefficient at the $k \mid N - k$ cut as $C_{(k)} = L_k R_k$, where the columns of L_k and rows of R_k are indexed by the cut bond. Thus every exact TT has bond at least $\text{rank } C_{(k)}$. For the converse, choose a basis of the column space of the first unfolding and express every remaining slice in that basis. Repeat on the residual coefficient array from left to right. At step k the number of retained basis vectors is exactly $\text{rank } C_{(k)}$, and after the final step their contracted product is C . This elementary rank-factorization construction is the exact TT-SVD argument [5]; it proves simultaneous attainability at every cut and not merely separate lower bounds. $\square$

Theorem 2.2 (Structural result II: universal exchange floor). *If $0 \neq C \in \Lambda^N V$, then*

$$\text{rank } C_{(k)} \geq \binom{N}{k}.$$

The lower bound is attained by every nonzero decomposable N -form.

Proof. Choose a nonzero coefficient c_I on an N -element support I . Restrict rows to k -subsets $A \subset I$ and columns to $I \setminus A$. The resulting $\binom{N}{k}$ square minor is diagonal up to signs: its diagonal entries are $\pm c_I$, while an off-diagonal entry repeats an index and vanishes. A decomposable form supported on an N -plane has no larger contraction image. $\square$

Corollary 2.3 (strict-antisymmetry truncation floor). *If $\tilde{C} \neq 0$ and $r_k^{\text{TT}}(\tilde{C}) < \binom{N}{k}$ at any cut, then $\tilde{C} \notin \Lambda^N V$.*

The TT-rank characterization is standard. The exterior rank-floor proof is included for audit, but no priority claim is made pending a dedicated exterior- algebra literature review. It is distinct from the stronger, generally open entropy-minimization statements studied for fermionic reduced density matrices [6, 4].

Theorem 2.4 (Structural result III: known flat Slater particle spectrum). *For orthonormal $u_1, \dots, u_N$, the normalized Slater determinant Φ has Schmidt decomposition*

$$\Phi = \frac{1}{\sqrt{\binom{N}{k}}} \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c}.$$

Thus all $\binom{N}{k}$ nonzero singular values equal $\binom{N}{k}^{-1/2}$. The best cut-rank- r approximation has relative squared error

$$\epsilon_{k,r}^2 = 1 - \frac{r}{\binom{N}{k}},$$

so error at most ϵ requires $r_k^{\text{TT}} \geq \lceil (1 - \epsilon^2) \binom{N}{k} \rceil$.

Proof. Partition the signed permutation sum by the set of orbitals in the first k slots. Each class factors into normalized internal Slater sums with coefficient $\sqrt{k!(N-k)!/N!}$. Distinct subsets give orthonormal states on both sides. The approximation formula is the Eckart–Young theorem applied to this flat singular-value list. This is the standard particle-reduced-density-matrix structure of a Slater determinant [7]. $\square$

No flat spectrum is asserted for a general correlated alternating tensor. Theorems 2.1, 2.2, and 2.4 concern labelled-particle TT only; they do not apply to mode MPS, ordered chambers, or exterior/determinant carriers.

For a fixed orthonormal one-particle basis of size D , all of these representations describe states in exactly the same finite physical space $\Lambda^N V_D$, whose full-configuration-interaction dimension is $\binom{D}{N}$. A mode-space FCI or quantum-chemistry DMRG calculation in that orbital basis changes the factorization and variational compression, not the physical finite-basis space [8, 2]. Consequently, FEMPS has neither a larger variational space nor an accuracy or efficiency advantage merely by being first quantized. Such an advantage would require matched basis, tolerance, timing, and memory comparisons.

3 Exterior matrix-product representation

Definition 3.1 (one-form matrix-wedge state). For site j , let $A^{[j]} \in \text{Mat}_{\chi_{j-1} \times \chi_j}(V_D)$ with $\chi_0 = \chi_N = 1$. For compatible matrices of forms define

$$(X \wedge Y)_{ac} = \sum_b X_{ab} \wedge Y_{bc}.$$

The open-boundary state is

$$C(A^{[1]}, \dots, A^{[N]}) = [A^{[1]} \wedge \dots \wedge A^{[N]}]_{11} \in \Lambda^N V_D. \quad (2)$$

Exterior associativity makes matrix-wedge multiplication associative, and each virtual path is a wedge of N one-forms. Hence the state is alternating by construction and $\epsilon_{\text{AS}} = 0$. The $\chi = 1$ family is exactly the decomposable forms. A sum of R Slaters embeds by R diagonal virtual paths, and the ordinary scalar MPS gauge action remains valid. These elementary facts do not supply a canonical form or an efficient norm contraction. The displayed parameter count $O(ND\chi^2)$ is an input-size statement only.

4 Fixed-bond exact contraction obstruction

For an $n \times n$ matrix $M = (M_{ij})$ over a noncommutative algebra, define the row-ordered Cayley determinant

$$\text{CDet}(M) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) M_{1,\sigma(1)} \cdots M_{n,\sigma(n)}.$$

Chien–Harsha–Sinclair–Srinivasan prove $\#P$ -hardness over $\text{Mat}_2(\mathbb{F})$ in characteristic zero [9]; related hardness results for the noncommutative determinant are given by Arvind–Srinivasan [10].

The distinction from exchange is essential. For scalar entries ($\chi = 1$), the factors commute, so the same alternating permutation sum is the ordinary determinant and is computable by Gaussian elimination; in FEMPS it is just the Slater exchange carrier. At $\chi = 2$, the entries lie in the fixed four-dimensional algebra $\text{Mat}_2(\mathbb{Q})$ and the product must remain in row order. The final physical wavefunction is still scalar and exactly antisymmetric: the noncommutativity lives only in the virtual computation.

Lemma 4.1 (direct Cayley coefficient). *Let $M_{ij} \in \text{Mat}_d(\mathbb{F})$ and set $\mathcal{A}^{[i]} = \sum_j M_{ij} e_j$. For $u, v \in \mathbb{F}^d$,*

$$u^\top \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[n]} v = (u^\top \text{CDet}(M) v) e_1 \wedge \cdots \wedge e_n.$$

After absorbing the boundaries into the endpoint cores, this is a FEMPS with $D = N = n$ and maximum internal bond d .

Proof. A nonzero top-form term chooses every physical basis vector once. The choice is a permutation, its exterior sign is the permutation sign, and its virtual matrix product remains in increasing site order. Their sum is $\text{CDet}(M)$. $\square$

Theorem 4.2 (Fixed-bond squared-norm hardness). *Over $\mathbb{Q}$, exact squared-norm evaluation for rational one-form FEMPS is $\#P$ -hard under a polynomial-time metric reduction even when $D = N$ and every internal bond is at most two. A generic exact contraction algorithm for fixed $\chi = 2$ would imply $\text{FP} = \#P$.*

Proof. Let φ be a 3CNF instance with m clauses and S satisfying assignments. Theorems 3.5 and 3.9 of Chien–Harsha–Sinclair–Srinivasan (CHSS) construct in polynomial time a polynomial-size matrix H_φ over $\text{Mat}_2(\mathbb{Q})$, with constant-bit integer gadget weights, such that

$$\text{CDet}(H_\varphi) = aI_2 + bJ_2, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad a + b = 4^{3m} S.$$

CHSS label the graph vertices, and their determinant is the row-ordered Cayley determinant in that vertex order. Apply Eq. (4.1) with $u = e_1$ and $v = e_1 + e_2$. The sole top-form coefficient is

$$u^\top (aI_2 + bJ_2) v = a + b = 4^{3m} S \geq 0,$$

so one exact squared-norm query returns $16^{3m} S^2$. Exact nonnegative integer square root followed by division by 4^{3m} recovers S . The CHSS graph size is polynomial in $|\varphi|$; all FEMPS cores have polynomial encoding length; and the queried norm and its square root have polynomial bit length because $S \leq 2^{n_{\text{var}}}$. Construction and postprocessing are therefore polynomial-time, and the maximum internal bond remains two. $\square$

The structured CHSS boundary is what sharpens the bond from three to two. For an arbitrary signed Cayley entry x , one may still adjoin the bond-one reference $\Theta = e_1 \wedge \cdots \wedge e_n$ by a block direct sum. The two norms x^2 and $(x+1)^2$ recover $x = ((x+1)^2 - x^2 - 1)/2$ at maximum bond three. That polarization observation is useful for general outputs, but it is not needed for the CHSS reduction.

CHSS work over any field of characteristic $p \neq 2$: their characteristic-zero statement is #P-hard and their positive odd-characteristic statement is Mod_pP -hard. We specialize to $\mathbb{Q}$ so that the input, output, square root, and bit-complexity model are explicit; no statement is made in characteristic two.

An independent integer verifier checks the site/physical permutation sum, all four boundary entries, the explicit bond-three direct sum, polarization, and $\epsilon_{\text{AS}} = 0$ for $2 \leq n \leq 6$. Its SHA-256 is `1d2208d3e5cb14f5c8e6c875f7fddf51c47ce9a3e61be6cedb8246d662b3a016`.

The result is not in conflict with the polynomial collapse of a fixed Mat_2 homogeneous pair power below. The hard one-form cores are site-dependent and preserve row order. Repeating one even two-form instead symmetrizes factor order.

The direct coefficient identity is pointwise algebraic and also holds when the entries are one-particle functions evaluated at a coordinate. A separate reduction for an explicit rational functional basis is proved in Theorem 12.1; it is independent of the norm reduction even though both start from the same CHSS source. Neither result says that every $\chi \geq 2$ FEMPS is unsuitable for Monte Carlo or controlled approximation.

4.1 Why parity is only part of the mechanism

The contrast above is not an odd-versus-even dichotomy. Site-dependent odd one-form cores retain the ordered virtual product and can encode the hard row-ordered determinant. Repetition of one homogeneous even two-form makes the physical factors commute and can collapse a fixed coefficient algebra. But scalar-bond odd one-form states are just Slater determinants and are easy, whereas the growing-width even sparse-path family below is hard. Parity explains the commutation step, while site dependence, homogeneity, virtual width, and algebraic closure decide the contraction problem. Splitting a general matrix-valued pair core into consecutive one-form cores can also increase the bond by a factor of order D ; it is not a cost-free equivalence.

4.2 Independent tagged pair-power mechanism

For $N = 2M$, a matrix-valued two-form $\Omega_B = \sum_{p < q} B_{pq} e_p \wedge e_q$ defines

$$\Psi_M(B; l, r) = l^\top \frac{\Omega_B^M}{M!} r.$$

Shift tags $E_{i,i+1} \otimes M_{ij}$ annihilate all factor orders except increasing row order. This gives a separate polynomial-time Turing reduction with pair bond $2(n+1)$ and one-form bond $O(N^2)$. It is weaker than Theorem 4.2 for unrestricted one-form cores, but isolates growing noncommutative order memory inside the homogeneous pair-power subfamily. Its order-1–4 exact-certificate hash is `893077be401414cd810fa1154e618d37d83b58e077732801f2482b3716b2c0c0`.

5 The matrix-pair corridor

5.1 Bounded Wedderburn–radical memory

Theorem 5.1 (Bounded-algebra LC–AGP collapse). *Let $\mathcal{A}$ be a finite-dimensional unital complex algebra with radical $R^d = 0$ and Wedderburn–Malcev decomposition*

$$\mathcal{A} = S \oplus R, \quad S = \bigoplus_{a=1}^q \text{Mat}_{p_a}(\mathbb{C}).$$

Put $p = \max p_a$ and $\rho = \dim R$. For any linear boundary functional λ , $\lambda(\Omega^M)/M!$ is an exact LC–AGP with

$$K \leq \sum_{k=0}^{\min(d-1, M)} q^{k+1} \rho^k \binom{M + v_k - 1}{v_k - 1}, \quad v_k = (k+1)p^2 + k.$$

For fixed p and d , this is polynomial in the explicit input dimensions. If all data, the displayed decomposition, and the boundary are supplied over $\mathbb{Q}$ with polynomial-bit structure constants, the LC-AGP coefficients can be constructed in deterministic polynomial Turing time and have polynomial bit length. No algorithm for discovering a rational Wedderburn–Malcev decomposition is asserted here.

Proof. Expand $\Omega = \Omega_S + \Omega_R$ into noncommutative words. A word with at least d radical factors vanishes. For a surviving number k , resolve the $k + 1$ semisimple runs through the q blocks and each radical insertion in a basis of R , giving at most $q^{k+1}\rho^k$ structural choices. For one choice, even physical two-forms commute; after applying λ , the result is a homogeneous degree- M scalar polynomial in at most $v_k = (k + 1)p^2 + k$ two-forms. In characteristic zero, M th powers of linear forms span $\text{Sym}^M(\mathbb{C}^{v_k})$. Each such power divided by $M!$ is one AGP, proving Eq. (5.1). The radical filtration, block expansion, and polarization count are written out in Appendix A. $\square$

The 2×2 upper-triangular and full Mat_2 base cases have exact certificates through $M = 6$ and $M = 4$, respectively. The latter needs at most $\binom{M+3}{3}$ AGPs even though the row-ordered determinant over the same matrix algebra is hard. This theorem classifies a pair-power family; it does not give an algorithm for the site-dependent cores of Theorem 4.2.

5.2 Selected growing memories

Theorem 5.2 (Fixed-state graded LC-AGP collapse). *Suppose the coefficient algebra embeds faithfully in*

$$\text{Mat}_w\left(\mathbb{C}[z_1, \dots, z_g]/(z_1^{d_1}, \dots, z_g^{d_g})\right),$$

with fixed w, g . Then every boundary contraction of $\Omega^M/M!$ is an exact LC-AGP with

$$K \leq \prod_{j=1}^g [M(d_j - 1) + 1] \binom{M + w^2 - 1}{w^2 - 1}.$$

In particular, the one-generator algebra $\mathbb{C}[z]/z^d$ requires at most $M(d - 1) + 1$ AGPs. For rational inputs with the embedding and quotient basis supplied explicitly, fixed w, g also gives a deterministic polynomial-time exact construction with polynomial-bit coefficients.

Proof. Before quotienting, every entry of the M th power has degree at most $M(d_j - 1)$ in counter z_j . Tensor-product Vandermonde interpolation extracts all retained coefficients using the first product in Eq. (5.2). At one grid point, a boundary of a $w \times w$ matrix power is a homogeneous degree- M polynomial in w^2 commuting two-forms; polarization supplies the second factor. Rational nodes give an exact rational construction for rational inputs; the quotient lift, interpolation matrices, reduction, and bit bounds are detailed in Appendix B. $\square$

The theorem includes a noncommutative alternating-word algebra of dimension $2d - 1$ embedded in $\text{Mat}_2(\mathbb{C}[z]/z^d)$. Thus growing nilpotency and two path branches remain insufficient when automaton width and counter count are fixed. Weighted-automata and Waring/Veronese ingredients are established prior art; the project-specific result is only their LC-AGP classification.

5.3 A bandwidth-one hard path

Let $P_j = e_{2j-1} \wedge e_{2j}$ be orthonormal pair forms. Given $A \in \mathbb{F}^{M \times M}$, define edge forms $F_i = \sum_j A_{ij} P_j$ on the unique endpoint path of an upper-bidiagonal $(M + 1) \times (M + 1)$ pair matrix.

Theorem 5.3 (Sparse APG permanent obstruction). *Over $\mathbb{Q}$ (or $\mathbb{R}$ with its positive inner product), for a zero-one matrix A the endpoint state and its squared norm are*

$$\Psi_A = \frac{F_1 \wedge \dots \wedge F_M}{M!} = \frac{\text{perm}(A)}{M!} P_1 \wedge \dots \wedge P_M, \quad \langle \Psi_A, \Psi_A \rangle = \frac{\text{perm}(A)^2}{(M!)^2}.$$

Exact squared-norm evaluation is therefore $\#P$ -hard for a family with pair bandwidth one, one virtual path, $D = 2M$, and $O(M^2)$ binary data. Over $\mathbb{C}$, the norm formula is $|\text{perm}(A)|^2/(M!)^2$ and the same zero-one reduction applies.

Proof. Repeated P_j factors vanish. Every surviving multilinear term selects a permutation, and even pair forms commute, so its coefficient is the permanent. For a zero-one matrix, multiply the exact norm by $(M!)^2$ and take the nonnegative integer square root; Valiant's permanent theorem applies [11]. $\square$

The state is an established antisymmetrized product of geminals, and permanent-valued APG/APIG coefficients and APG-to-AGP decompositions have direct prior art [12, 13]. The hard output is projectively a single top-form Slater, so this is an input-to-coefficient obstruction, not a lower bound on physical LC-AGP rank.

6 Relative approximation and energy certification

Exact permanent hardness alone does not exclude approximation. In particular, entrywise-nonnegative permanents admit an FPRAS [14]. The same sparse-path identity also accepts a different promise.

Theorem 6.1 (Real-PSD relative-norm transfer). *Unless $\text{RP} = \text{NP}$, there is no randomized polynomial-time relative approximation scheme for the squared norm of every rational real upper-bidiagonal matrix-pair input, polynomial jointly in dimensions, input bit length, inverse tolerance, and logarithmic inverse failure probability.*

Proof. For rational real positive-semidefinite A , the sparse-path construction gives

$$s_A = \frac{\text{perm}(A)^2}{(M!)^2}, \quad \text{perm}(A) \geq 0.$$

A relative estimate $\tilde{s}_A$ would make $M!\sqrt{\tilde{s}_A}$ a relative permanent estimate after a polynomial tolerance conversion. Meiburg excludes a PRAS for real-PSD permanents unless $\text{RP} = \text{NP}$ [15]. $\square$

Additive permanent estimators remain possible [16]. They do not uniformly imply relative norm control: cancellations can make the true coefficient exponentially small in input precision while the additive scale remains order one.

Lemma 6.2 (Rayleigh-quotient certificate). *If $n > 0$, $E = h/n$, and estimates obey*

$$|n - \tilde{n}| \leq \Delta_n, \quad |h - \tilde{h}| \leq \Delta_h, \quad \tilde{n} > \Delta_n,$$

then, with $\tilde{E} = \tilde{h}/\tilde{n}$,

$$|E - \tilde{E}| \leq \frac{\Delta_h + |\tilde{E}|\Delta_n}{\tilde{n} - \Delta_n}.$$

For random bounds this holds on the simultaneous success event.

Proof. Use $E - \tilde{E} = [(h - \tilde{h}) - \tilde{E}(n - \tilde{n})]/n$ and $n \geq \tilde{n} - \Delta_n > 0$. $\square$

Thus estimator unbiasedness alone is not an energy certificate; the denominator interval must stay positive. Tensor-network Monte Carlo [17] and sign-problem complexity [18] provide context but are not used as the reduction.

7 Failure of a universal direct statistics carrier

For $0 \neq C \in \Lambda^N V$, define the cut contraction

$$c_{C,k} : \Lambda^k V^* \longrightarrow \Lambda^{N-k} V, \quad c_{C,k}(\alpha) = \iota_\alpha C,$$

and $\mathcal{B}_k(C) = \text{im } c_{C,k}$. The tested proposal was an exact universal factorization

$$\mathcal{B}_k(C) \cong \mathcal{S}_{N,k}^{\text{fermion}} \otimes \mathcal{M}_k(C), \quad \dim \mathcal{M}_k(\text{Slater}) = 1. \quad (3)$$

Theorem 7.1 (Cut-rank divisibility obstruction). *For every $N \geq 3$, Eq. (3) cannot hold for every nonzero N -form.*

Proof. At the $1 \mid N-1$ cut, a Slater determinant has rank N , forcing $\dim \mathcal{S}_{N,1}^{\text{fermion}} = N$. Hence every admitted rank would be divisible by N . In $V = \mathbb{F}^{N+2}$, let

$$C_N = e_1 \wedge (e_2 \wedge e_3 + e_4 \wedge e_5) \wedge e_6 \wedge \cdots \wedge e_{N+2}.$$

Its $N+2$ one-index contractions have distinct monomial supports and are linearly independent, so $\text{rank } c_{C_N,1} = N+2$. But $N \nmid N+2$. $\square$

The counterexample is only two Slaters. Shared physical covectors lock $N-2$ channels that an independent multiplicity tensor would treat as free. The rank is invariant under orbital basis changes and direct embeddings, and full one-cut rank is stable on an open neighborhood.

Standard symmetry-adapted tensor networks remain valid [19, 20]. Particle permutation supplies the one-dimensional sign representation, not a $\binom{N}{k}$ degeneracy. Under $GL(V)$, $\Lambda^N V$ is irreducible and the exterior cut has Littlewood–Richardson multiplicity one while retaining the full orbital representation. Hamiltonian-specific charges and multiplets are not ruled out. State-adaptive Slater/secant descriptions are established and need not be canonical [21, 22, 23].

The sparse-path construction sharpens the algorithmic warning: whenever its output is nonzero it is projectively one Slater, yet its scale is a permanent. A state-intrinsic multiplicity-one label cannot by itself compile compact cores into a norm algorithm.

8 Established positive controls and prior-art boundary

Fixed-number AGP is equivalent to number-projected pairing structure; projected BCS, HFB overlaps, and Pfaffian formulas are established [24, 25, 26]. AGP norms and reduced density matrices have polynomial algorithms [27]. Deterministic linear combinations of optimized and nonorthogonal AGPs precede this project [28, 29]; Pfaffian variational Monte Carlo with symmetry projection is also established [30, 31].

Accordingly, finite LC-AGP is used here as a validated tractable control, not as a new ansatz. The two collapse theorems above classify when matrix-pair memory reduces to that control. Grassmann signs or parity bookkeeping enforce fermionic algebra but do not, without additional closure, refute either hardness construction.

Lemma 8.1 (Explicit AGP embedding in one-form FEMPS). *Let a scalar two-form be in skew-normal form $\Omega_F = \sum_{a=1}^r w_a u_a \wedge v_a$. For $1 \leq M \leq r$,*

$$\frac{\Omega_F^M}{M!} = \sum_{a_1 < \cdots < a_M} \left(\prod_{j=1}^M w_{a_j} \right) u_{a_1} \wedge v_{a_1} \wedge \cdots \wedge u_{a_M} \wedge v_{a_M}$$

has an exact one-form FEMPS with every internal bond at most $r \leq \lfloor D/2 \rfloor$.

Proof. Use the channel label a as the virtual state. The first core emits $w_a u_a$, the next core emits v_a diagonally, and every later u core has only strictly upper-triangular transitions $a < b$ with entry $w_b u_b$; the following v core is diagonal. Contracting the final v core with the right boundary leaves exactly the strictly increasing channel sequences displayed above. Thus the construction has $\binom{r}{M}$ nonzero paths but bond r , and never enumerates the paths in its core representation. $\square$

For odd particle number, a blocked orbital b gives $b \wedge \Omega_F^M / M!$ by prepending one bond-one core; no auxiliary subtraction is needed for this representation statement. This embedding is a prior-art control, not a new AGP ansatz. It also does not contradict the $O(\chi D)$ bond that can arise when a general matrix-valued pair core of bond χ is split into one-form sites: skew-normal scalar AGP structure is the special closure used by the lemma.

9 Numerical validation and algorithm-design consequences

The structural results were subsequently used to choose, rather than reject, a restricted numerical control. A nonbranching diagonal-path subclass is exactly a K -term nonorthogonal Slater expansion and evaluates determinant transition quantities in $O(K^2)$ path pairs without enumerating virtual histories or materializing the alternating coefficient tensor. This places it within established NOCI methodology rather than defining a new correlated ansatz [32, 33].

We retain one representative interacting result. For six one-dimensional fermions with harmonic confinement and softened Coulomb repulsion, the $D = 12$, $K = 4$ calculation occupies the $\binom{12}{6} = 924$ dimensional exterior space and gives energy 25.0494711446, compared with same-basis CI 25.0493664161. The error is 1.0473×10^{-4} , the energy variance is 1.1223×10^{-3} , the norm error is 1.55×10^{-15} , and the structural antisymmetry residual is exactly zero. The run took 107.7 s on CPU and its sampled peak resident memory was 0.975 GB. The $D = 12$, $K = 4$ orbitals were *not* initialized from a clean $D = 12$ Slater: they were obtained by exact zero-padding of a preoptimized $D = 10$, $K = 4$ state before the $D = 12$ optimization. Direct CI was used only as an independent reference.

These floating-point data are a bounded NOCI-equivalent numerical exercise. They do not establish an ansatz beyond NOCI, an explicit-correlation advantage, improved functional-basis D convergence, asymptotic scalability, or superiority over FCI, particle TT, Li–Waintal, or same-basis DMRG.

10 Logical boundaries of the results

Table 1 records transitions most likely to create an invalid claim. Every row is a manuscript constraint, not an empirical observation.

11 Reproducibility checks

Table 2 lists bounded checks. Each verifier imports neither the production tensor backend nor the statement it checks. Passing a bounded certificate is evidence against transcription errors; it is not a proof for arbitrary size.

The complete certificate suite is reproduced from the repository root by

```
python -m pytest tests/test_exact_certificates.py -q
```

The test module records every verifier–certificate pair and invokes each standalone verifier in audit mode; Table 2 records the independently enumerated range for every claim.

The corresponding full SHA-256 values are:

Transition	What is established	What is not inferred
Exact/approximate	The two hardness constructions use exact arithmetic; the relative-norm theorem separately treats a randomized scheme.	Exact #P-hardness alone does not rule out an FPRAS or additive estimator.
Affine/projective	Norm polarization recovers the signed affine scale; the sparse-path output is projectively one Slater.	A simple normalized state does not make its compact-input normalization easy.
State/input	All complexity bounds measure explicit rational cores and coefficient bit length.	Minimal intrinsic Slater rank is not an input decoder or contraction algorithm.
Representation	The exchange-rank results apply to labelled-particle TT; fixed-bond hardness applies to exterior one-form cores.	Neither theorem transfers automatically to mode MPS, ordered sectors, or every fermionic TN.
Homogeneous/site-dependent	The collapse results repeat one matrix-valued even form; the Cayley construction uses row-labelled one-form cores.	Fixed Mat_2 pair-power collapse does not make fixed-bond site-dependent Cayley cores easy.
Energy/norm	A simultaneous numerator/norm event with a positive denominator interval certifies energy.	Unbiased numerator and norm estimates do not make their ratio unbiased or certified.

Table 1: Mandatory proof and wording transitions.

- Fixed-bond norm:
1d2208d3e5cb14f5c8e6c875f7fddf51c47ce9a3e61be6cedb8246d662b3a016.
- Point evaluation verifier:
f2ed7656126cdb408a312b97f6ac9a8cbf0f323896ef14919d3c67998456b4aa.
- Tagged pair power:
893077be401414cd810fa1154e618d37d83b58e077732801f2482b3716b2c0c0.
- Bounded algebra, T_2 :
f671c2c10376c39cfb8c223edafba370570b9c417e8b12bcaaeb2f0f66cf078c.
 Mat_2 :
74d2de4a2cedcbaf548cd4c9895d0ea4af48e6bb485b72966562e46277a6d20d.
- Graded algebra, one counter:
07a222de3f44ced1b3fe155638299fdb8443a54f3d9f36479b994f32f9f0fd55.
Alternating words:
082238ff6e6783b7533b3b2a59f3664beb1820794bcae573add98baf43370030.
- Sparse path:
dd72c1aaeb0bc2a6b9206992cde9099f2f568b7ff6c8ed8eb7e38d958f78e790.
- Relative norm:
c15e7ff268a962e2790004c7f63d47bedb53be0c887ab0241e671b7fe4ff3b16.
- Carrier obstruction:
06025f168a49c0ab857c2163103ffabcb56fb04cd1fed4df9120d25ef6bc60df.

Claim	Range	What the verifier checks
Fixed-bond norm	$2 \leq N \leq 6$	Direct Cayley coefficient, four generic boundaries, explicit bond-three polarization, and exact antisymmetry; the bond-two hardness step uses the published CHSS structured output.
Point evaluation	$2 \leq N \leq 6$	Shifted-Legendre evaluation determinant and inverse, and the direct Cayley point formula of Eq. (10).
Tagged pair power	orders 1–4	Perfect matchings and every pair-factor order against the tagged row-ordered determinant.
Bounded algebra	T_2 : $M \leq 6$; Mat_2 : $M \leq 4$	Exact polynomial powers versus simplex interpolation and LC-AGP coefficients.
Graded algebra	$M, d \leq 4$; alternating words $M \leq 3, d \leq 4$	Every boundary coefficient, Vandermonde extraction, and nested matrix-power interpolation.
Sparse path	$1 \leq M \leq 6$	Unique path, exterior subset expansion, and permutation permanent route.
Relative norm	bounded rational cases	Positive, cancelling, precision-ill-conditioned, signed PSD, and Rayleigh interval controls.
Carrier obstruction	$3 \leq N \leq 8$	All cut ranks, channel locking, orbital permutation, embedding, and full-rank perturbation.

Table 2: Exact-certificate coverage.

12 Discussion: a two-mechanism map for antisymmetrization

Consider exact evaluation at a fixed configuration $x = (x_1, \dots, x_N)$ of

$$(\mathcal{A}F)(x) = \sum_{\pi \in S_N} \text{sgn}(\pi) F(x_{\pi(1)}, \dots, x_{\pi(N)}). \quad (4)$$

Two independent properties organize several familiar constructions. Property P1 (*assignment factorization*) asks whether, after fixing x , every permutation term has the form

$$F_{\pi}(x) = \prod_{i=1}^N a_{i, \pi(i)}(x), \quad (5)$$

where the permutation selects factors only through $(i, \pi(i))$. The factors $a_{ij}(x)$ may depend on the *entire* configuration, as in backflow or FermiNet; P1 is not a one-particle locality assumption. Property P2 asks whether these factors multiply in one commutative coefficient algebra. In every cell below we additionally assume that each factor is itself evaluable from its encoded input in polynomial time and polynomial bit complexity. Without this base assumption, an easy determinant layer need not make the complete model easy.

In the upper-left cell, the ordinary Slater determinant replaces an explicit $N!$ -term Leibniz sum by one determinant. At fixed x , generalized Slater, backflow, and FermiNet orbital entries remain scalars in a commutative field, even when every entry depends on all electronic coordinates; permutation equivariance makes particle exchange act by row or column exchange. Each determinant channel is therefore polynomial-time once its entries are available. Gaussian elimination applies over fields. Over a general commutative ring, divisions need not exist, but division-free determinant algorithms such as Berkowitz’s still give polynomial-size arithmetic circuits [34]. Hutter’s generalized-Slater universality analysis, FermiNet, the explicitly antisymmetrized layers of Lin–Goldshlager–Lin, and the separation of Zweig–Bruna concern representation, universality, ansatz design, or expressivity [35, 36, 37, 38]; they are not attributed the complexity map in Table 3.

The upper-right cell isolates the second mechanism. In one-form FEMPS, Eq. (5) holds with matrix-valued factors, but row order prevents their exchange. The exterior algebra supplies

	P2: commutative	not P2: noncommutative
P1	Slater, generalized Slater, backflow, and FermiNet determinant channels: the $N!$ Leibniz terms collapse to ordinary determinants.	One-form FEMPS: exterior signs accompany row-ordered virtual products, giving a Cayley determinant.
not P1	The signed Hamilton-path polynomial in Eq. (11): open cell in the present bounded search.	General explicit antisymmetrization; conditional hardness only for a precisely defined class that exactly contains the hard bond-two one-form cores with polynomial overhead.

Table 3: A two-mechanism map, not a complete dichotomy or classification theorem. P1 concerns assignment structure; P2 concerns the coefficient algebra.

$\text{sgn}(\pi)$ while the virtual algebra retains the noncommuting product, producing the row-ordered Cayley determinant rather than a scalar determinant. The following point-evaluation theorem makes this distinction explicit without changing the squared-norm theorem above.

Theorem 12.1 (Exact rational-polynomial point evaluation). *Let $\ell_r(t) = P_r(2t - 1)$, $r \geq 0$, be the unnormalized shifted Legendre polynomials, where $P_r(1) = 1$. Exact pointwise evaluation of rational one-form FEMPS represented in the basis $\ell_0, \dots, \ell_{D-1}$ is $\#P$ -hard under polynomial-time metric reductions, already when $D = N$ and every internal bond is at most two. For the isometric convention J_N , the exact output is encoded as the rational q in $\Psi(x) = q/\sqrt{N!}$; equivalently one may request the unnormalized rational alternating value q .*

Proof. The shifted polynomials have the integer expansion

$$\ell_r(t) = \sum_{k=0}^r (-1)^{r-k} \binom{r}{k} \binom{r+k}{k} t^k \quad (6)$$

and leading coefficient $\binom{2r}{r}$. Put $t_i = i/(N+1)$ and $B_{ij} = \ell_{j-1}(t_i)$. Degree grading and the Vandermonde determinant give

$$\det B = \left(\prod_{r=0}^{N-1} \binom{2r}{r} \right) \prod_{1 \leq i < k \leq N} (t_k - t_i) \neq 0. \quad (7)$$

The nodes use $O(\log N)$ bits. Equation (6) shows that all evaluations use polynomially many bits. More explicitly, $Q = (N+1)^{N-1}$ clears every denominator in B ; the entries of QB have polynomial bit length, and Hadamard's bound gives polynomial bit length for its determinant and all minors. Thus $\det B$, $(\det B)^{-1}$, and $B^{-1} = Q \text{adj}(QB)/\det(QB)$ have polynomial bit length and are computable by exact rational arithmetic in polynomial time.

For a 3CNF formula φ , take the CHSS matrix $H_\varphi = (H_{ij}) \in \text{Mat}_N(\text{Mat}_2(\mathbb{Q}))$ from the proof of Theorem 4.2, and define the functional cores

$$\mathcal{A}^{[i]}(t) = \sum_{j=1}^N H_{ij} \ell_{j-1}(t), \quad u = e_1, \quad v = e_1 + e_2. \quad (8)$$

This mapping merely reuses the rational CHSS entries as functional-basis coefficients, so its size and coefficient bit length are polynomial. Expanding the alternating point value at $(t_1, \dots, t_N)$, and using commutativity only for the scalar B_{ij} , gives

$$\Psi(t_1, \dots, t_N) = \frac{\det B}{\sqrt{N!}} u^\top \text{CDet}(H_\varphi) v \quad (9)$$

$$= \frac{\det B}{\sqrt{N!}}, 4^{3m} \# \text{SAT}(\varphi). \quad (10)$$

Indeed, a repeated basis-column choice vanishes, while a column permutation σ contributes $\text{sgn}(\sigma) \det B$ and leaves the row-ordered product $H_{1\sigma(1)} \cdots H_{N\sigma(N)}$. Thus the known nonzero factor is $c_N = 1/\sqrt{N!}$ for J_N and $c_N = 1$ for the unnormalized antisymmetrizer. The normalized output convention stores the exact algebraic number as (q, N) meaning $q/\sqrt{N!}$; $N!$ has $O(N \log N)$ bits, so this is an explicit polynomial-size output domain rather than a hidden algebraic-number oracle. One query, followed by rational division by $4^{3m} \det B$, recovers $\#\text{SAT}(\varphi)$. This is a polynomial-time metric reduction. $\square$

Theorem 12.1 and Theorem 4.2 are independent reductions from the same Cayley source; neither is inferred from the other. They concern worst-case exact evaluation and do not preclude efficient approximate QMC/VMC, promised subclasses, or particular low-bond instances.

For the lower-left cell, the exact path object is

$$H_{\text{sgn}}(W) = \sum_{\pi \in S_N} \text{sgn}(\pi) \prod_{i=1}^{N-1} w_{\pi(i), \pi(i+1)}. \quad (11)$$

Its scalar factors commute, but adjacent vertices couple successive permutation positions, so it need not have P1. This is a path, not a closed cycle: summing a cycle over all permutations would cancel for even N because a cyclic shift reverses permutation parity. Hamilton path/cycle polynomials, sign imbalance, even-minus-odd path counts, fermionants, and immanants were checked in a bounded search; the latter neighboring families have their own complexity results [39]. In the present bounded literature search, the authors did not find an existing complexity classification for this exact weighted polynomial. We therefore register the cell as open and make no hardness or tractability claim.

The lower-right entry supports only a conditional inheritance statement. If a precisely defined parameterized function class exactly represents every hard bond-two instance in Theorem 12.1 with polynomial overhead, then its exact point-evaluation problem inherits that $\#\text{P}$ -hardness. Universal approximation does not imply exact containment. In particular, Lin–Goldshlager–Lin motivate generic explicit antisymmetrization, including its factorial sum, but no exact containment lemma is asserted here for their neural class.

Finally, if J is fully permutation-symmetric, then directly from Eq. (4),

$$J \mathcal{A}(M) = \mathcal{A}(JM). \quad (12)$$

If J has a compatible scalar particle-TT representation of bond χ_J and multiplication closes in the chosen functional basis or in a stated finite extension, tensoring its virtual indices with a Slater carrier gives a conditional Slater–Jastrow embedding with the corresponding χ_J bond factor. This is an embedding statement, not a norm-hardness reduction; the point value remains $J(x) \det \Phi(x)$ up to normalization. The Zweig–Bruna separation concerns expressive efficiency relative to finite Slater sums, not point- or norm-evaluation complexity.

The table is an organizing framework and partial classification program. A complete complexity classification of structured antisymmetrization, including the unresolved cells and internal tractability boundaries, is left for future work.

12.1 Limitations and next questions

The results do not establish that every exterior family is hard, that every tractable exterior family is LC–AGP, or that additive approximation with a certified polynomial norm lower bound is impossible. Promised positive, low-rank, strongly orthogonal, Pfaffian/Gaussian, integrable, or low-treewidth families remain admissible if their promise excludes the hard embeddings and their state and energy errors are certified. A genuinely different categorical carrier is not excluded by the divisibility obstruction; it must state its exactness, reconstruction cost, and contraction theorem rather than assume a free direct product.

The structural results and all currently admitted restricted-solver numerics belong to this single manuscript. No separate FEMPS method paper is claimed from the diagonal-path data. A future method manuscript is opened only after a representation beyond finite NOCI demonstrates a matched, reproducible advantage—for example, explicit correlation that improves functional-basis D convergence, or a controlled comparison with Li–Waintal and same-basis DMRG that identifies a genuine accuracy, stability, or complexity tradeoff. Higher-dimensional alternating-form rank classification is not pursued unless a specific rank question determines such an algorithmic or physical decision.

OpenAI Codex (GPT-5 family, accessed in August–September 2026) was used under human direction for literature discovery, code and proof exploration, reproducibility checks, and language editing. The human author selected the research questions, inspected the cited sources, verified the symbolic and numerical outputs, revised every proof and claim, and accepts responsibility for the manuscript. The AI system is not an author.

13 Conclusion

The attempted fermionic completion of the 2201 functional tensor network has identified two costs that must not be conflated. Ordinary particle TT pays a binomial exchange bond even for a Slater determinant. Exteriorizing the state removes that representation charge, but generic exact squared-norm contraction and rational-polynomial point evaluation are already hard at fixed one-form bond, while a direct universal carrier factorization fails dimensionally. The tested tractable matrix-pair memories reduce to established LC–AGP structure, and a unique sparse path already contains a permanent. The nonbranching K -path calculation is correspondingly a useful NOCI-equivalent numerical control, not yet an independent method contribution. The resulting map is an algorithm-design constraint, not a denial of numerical FEMPS: genuinely distinct restricted subclasses and error-controlled approximate methods remain open and must be judged by matched interacting benchmarks, D and correlation convergence, cost, uncertainty, and antisymmetry residuals rather than by parameter counts alone.

A Detailed bounded-algebra collapse proof

This appendix expands the algebraic steps used in the bounded-algebra collapse theorem. Let $E = \Lambda^{\text{even}} V_D$. The wedge product makes E a commutative $\mathbb{C}$ -algebra, although it has zero divisors. Write $\Omega = \Omega_S + \Omega_R \in \mathcal{A} \otimes E_2$, where $E_2 = \Lambda^2 V_D$.

Radical filtration. Because R is a two-sided ideal, a product containing k factors from R , with arbitrary elements of S between them, belongs to R^k . This follows inductively: $SR^j, R^jS \subseteq R^j$, and $R^iR^j \subseteq R^{i+j}$. Hence every word in the expansion of Ω^M with $k \geq d$ radical factors vanishes. For fixed $k < d$, every surviving word has a unique run-length description

$$\Omega_S^{m_0} \Omega_R \Omega_S^{m_1} \cdots \Omega_R \Omega_S^{m_k}, \quad m_0 + \cdots + m_k = M - k,$$

where zero-length semisimple runs are allowed. Summing over all nonnegative run lengths is equivalent to retaining the homogeneous degree- M part of the commutative physical-form polynomial constructed below; no additional combinatorial factor is omitted because the virtual product order remains the displayed order.

Wedderburn blocks and radical basis. Let 1_a be the central identity of the block $S_a \cong \text{Mat}_{p_a}(\mathbb{C})$ and choose a basis $r_1, \dots, r_\rho$ of R . Insert $1_S = \sum_{a=1}^q 1_a$ independently into each of

the $k + 1$ semisimple runs and expand each radical factor as

$$\Omega_R = \sum_{b=1}^{\rho} r_b \eta_b, \quad \eta_b \in E_2.$$

Thus the contribution with k radical occurrences is a sum over at most $q^{k+1}\rho^k$ choices $\tau = (a_0, b_1, a_1, \dots, b_k, a_k)$. This is an upper bound: incompatible block transitions may vanish and repeated basis choices may combine.

Fix one τ . In matrix units of S_{a_j} , the corresponding projected semisimple form has at most $p_{a_j}^2 \leq p^2$ scalar two-form entries $x_{j,\mu\nu} \in E_2$. Together with the selected radical forms $\eta_{b_1}, \dots, \eta_{b_k}$, the boundary value under λ is a homogeneous degree- M polynomial

$$P_\tau(x_{0,\mu\nu}, \dots, x_{k,\mu\nu}, \eta_{b_1}, \dots, \eta_{b_k}) \in E_{2M}$$

in at most $v_k = \sum_{j=0}^k p_{a_j}^2 + k \leq (k+1)p^2 + k$ commuting two-forms. Matrix multiplication and λ affect only the scalar coefficients of this polynomial. Nilpotence inside E can identify or kill monomials, but cannot increase the number of powers needed below.

Polarization into AGPs. Let $z_1, \dots, z_v$ be formal commuting variables. Over characteristic zero, put $T_M = \{t \in \mathbb{Z}_{\geq 0}^{v-1} : |t| \leq M\}$. The integer simplex is unisolvant for polynomials of total degree at most M : ordered by total degree, the falling-factorial polynomials $\prod_{i=2}^v \binom{x_i}{\alpha_i}$ give a triangular evaluation matrix on T_M with diagonal one. Hence the square matrix relating the $|T_M| = \binom{M+v-1}{v-1}$ powers

$$(z_1 + t_2 z_2 + \dots + t_v z_v)^M, \quad t \in T_M,$$

to the degree- M monomial basis is nonsingular (the multinomial factors are nonzero in characteristic zero). Therefore every homogeneous degree- M polynomial in v variables is a linear combination of at most $|T_M|$ such powers.

For rational data this is also a Turing construction. The evaluation entries are integers of bit length $O(M \log(M+1))$; for fixed v the matrix order is $O(M^{v-1})$. Hadamard's inequality and Cramer's rule bound every numerator and denominator in its inverse by a polynomial number of bits. Exact Gaussian elimination therefore computes the power coefficients in polynomial time. In the theorem, $v \leq v_k$ is fixed when p, d are fixed; the supplied rational block decomposition and radical basis have polynomial-bit entries, and the number of structural choices is polynomial. Multiplication, application of λ , and division by $M!$ increase bit lengths only polynomially.

Substitute the scalar two-forms for the z_i and apply the algebra homomorphism from the symmetric algebra to E . Every resulting term $L^M/M!$ is an AGP. For a fixed k , multiplying the power count by the at most $q^{k+1}\rho^k$ structural choices and summing over $0 \leq k \leq \min(d-1, M)$ gives Eq. (5.1). All steps use only the fixed Wedderburn decomposition, radical basis, matrix multiplication, and exact linear algebra. For fixed p and d , the displayed count is polynomial in M, q , and ρ , hence in the explicit algebraic input size.

B Detailed fixed-state graded collapse proof

Let $R_d = \mathbb{Q}[z_1, \dots, z_g]/(z_1^{d_1}, \dots, z_g^{d_g})$ and suppose the embedding into $\text{Mat}_w(R_d)$ is part of the input. Represent every residue by its canonical polynomial lift with $0 \leq \deg_{z_j} < d_j$. An entry of the unreduced product Ω^M then has $\deg_{z_j} \leq L_j = M(d_j - 1)$. For each counter use the rational grid $0, 1, \dots, L_j$. The univariate Vandermonde matrix is nonsingular, and the tensor product of these matrices recovers every coefficient of the lifted product. Reducing modulo $z_j^{d_j}$ simply discards coefficients whose z_j exponent is at least d_j ; it cannot create additional terms.

At one grid point, Ω is a $w \times w$ matrix whose w^2 entries are commuting physical two-forms. After the fixed boundary is applied, Ω^M is a homogeneous degree- M polynomial in at most w^2

such forms. The integer-simplex polarization construction in Appendix A expresses it as at most $\binom{M+w^2-1}{w^2-1}$ powers. Recombining the at most $\prod_j (L_j + 1)$ grid contributions gives Eq. (5.2).

For the bit model, every Vandermonde entry has polynomial bit length in L_j ; Hadamard’s inequality and Cramer’s rule give polynomial bit length for its inverse. Because g is fixed and the quotient basis of dimension $\prod_j d_j$ is explicit, $\prod_j (L_j + 1)$ is polynomial in the stated input and M . The polarization dimension w^2 is fixed. Exact rational matrix multiplication, tensor-product interpolation, quotient reduction, boundary application, and division by $M!$ therefore use polynomially many bit operations and produce polynomial-bit LC–AGP coefficients. This conclusion requires the rational embedding and quotient representation as input; it does not claim an algorithm for discovering such an embedding from arbitrary structure constants.

References

- [1] Rui Hong, Ya-Xuan Xiao, Jie Hu, An-Chun Ji, and Shi-Ju Ran. Functional tensor network solving many-body schrödinger equation. *Physical Review B*, 105:165116, 2022.
- [2] Zhendong Li and Garnet Kin-Lic Chan. Hilbert space renormalization for the many-electron problem. *The Journal of Chemical Physics*, 144:084103, 2016.
- [3] Jian-Gang Kong, Jia-Ji Zhu, and Z. Y. Xie. Grassmann tensor networks. *Annals of Physics*, 491:170511, 2026.
- [4] Jing Li and Xavier Waintal. Matrix product states and first quantization. *Physical Review Letters*, 136:116503, 2026.
- [5] Ivan V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33:2295–2317, 2011.
- [6] Eric A. Carlen, Elliott H. Lieb, and Robin Reuvers. Entropy and entanglement bounds for reduced density matrices of fermionic states. *Communications in Mathematical Physics*, 344(3):655–671, 2016.
- [7] A. John Coleman. Structure of fermion density matrices. *Reviews of Modern Physics*, 35:668–686, 1963.
- [8] Garnet Kin-Lic Chan and Sandeep Sharma. The density matrix renormalization group in quantum chemistry. *Annual Review of Physical Chemistry*, 62:465–481, 2011.
- [9] Steve Chien, Prahladh Harsha, Alistair Sinclair, and Srikanth Srinivasan. Almost settling the hardness of noncommutative determinant. In *Proceedings of the 43rd ACM Symposium on Theory of Computing*, pages 499–508, 2011.
- [10] V. Arvind and Srikanth Srinivasan. On the hardness of the noncommutative determinant. *Computational Complexity*, 27:1–29, 2018.
- [11] Leslie G. Valiant. The complexity of computing the permanent. *Theoretical Computer Science*, 8(2):189–201, 1979.
- [12] Airi Kawasaki and Naoki Nakatani. Low-rank decomposition on the antisymmetric product of geminals for strongly correlated electrons, 2024. preprint.
- [13] Michelle Richer, Taewon D. Kim, and Paul W. Ayers. Graphical approach to interpreting and efficiently evaluating geminal wavefunctions. *International Journal of Quantum Chemistry*, 125(1):e70000, 2025.
- [14] Mark Jerrum, Alistair Sinclair, and Eric Vigoda. A polynomial-time approximation algorithm for the permanent of a matrix with nonnegative entries. *Journal of the ACM*, 51(4):671–697, 2004.
- [15] Alexander Meiburg. Inapproximability of positive semidefinite permanents and quantum state tomography. *Algorithmica*, 85:3828–3854, 2023.
- [16] Scott Aaronson and Travis Hance. Generalizing and derandomizing gurvits’s approximation algorithm for the permanent. *Quantum Information and Computation*, 14(7–8):541–559, 2014.
- [17] Andrew J. Ferris. Unbiased monte carlo for the age of tensor networks, 2015. preprint.
- [18] Matthias Troyer and Uwe-Jens Wiese. Computational complexity and fundamental limitations to fermionic quantum monte carlo simulations. *Physical Review Letters*, 94:170201, 2005.

- [19] Sukhwinder Singh, Robert N. C. Pfeifer, and Guifré Vidal. Tensor network decompositions in the presence of a global symmetry. *Physical Review A*, 82:050301, 2010.
- [20] Andreas Weichselbaum. Non-abelian symmetries in tensor networks: A quantum symmetry space approach. *Annals of Physics*, 327:2972–3047, 2012.
- [21] John Schliemann, J. Ignacio Cirac, Marek Kuś, Maciej Lewenstein, and Daniel Loss. Quantum correlations in two-fermion systems. *Physical Review A*, 64:022303, 2001.
- [22] Edoardo Ballico, Alessandra Bernardi, Maria Virginia Catalisano, and Luca Chiantini. Grassmann secants, identifiability, and linear systems of tensors. *Linear Algebra and its Applications*, 438:121–135, 2013.
- [23] Vincenzo Galgano and Reynaldo Staffolani. Identifiability and singular locus of secant varieties to grassmannians. *Collectanea Mathematica*, 2024.
- [24] Jorge Dukelsky, Stuart Pittel, and Carlos Esebbag. Structure of the number projected BCS wave function, 2016. preprint.
- [25] Luis M. Robledo. Sign of the overlap of Hartree–Fock–Bogoliubov wave functions. *Physical Review C*, 79:021302, 2009.
- [26] G. F. Bertsch and L. M. Robledo. Symmetry restoration in hartree–fock–bogoliubov based theories. *Physical Review Letters*, 108:042505, 2012.
- [27] Armin Khamoshi, Thomas M. Henderson, and Gustavo E. Scuseria. Efficient evaluation of AGP reduced density matrices. *The Journal of Chemical Physics*, 151:184103, 2019.
- [28] Wataru Uemura, Shusuke Kasamatsu, and Osamu Sugino. Configuration interaction with antisymmetrized geminal powers. *Physical Review A*, 91:062504, 2015.
- [29] Rishab Dutta, Guo P. Chen, Thomas M. Henderson, and Gustavo E. Scuseria. Construction of linearly independent non-orthogonal AGP states. *The Journal of Chemical Physics*, 154:114112, 2021.
- [30] Daisuke Tahara and Masatoshi Imada. Variational monte carlo method combined with quantum-number projection and multi-variable optimization. *Journal of the Physical Society of Japan*, 77:114701, 2008.
- [31] Michal Bajdich, Lubos Mitas, Lucas K. Wagner, and Kevin E. Schmidt. Pfaffian pairing and backflow wave functions for electronic structure quantum monte carlo methods. *Physical Review B*, 77:115112, 2008.
- [32] Alex J. W. Thom and Martin Head-Gordon. Hartree–fock solutions as a quasidiabatic basis for nonorthogonal configuration interaction. *The Journal of Chemical Physics*, 131:124113, 2009.
- [33] Gregory Beylkin, Martin J. Mohlenkamp, and Fernando Pérez. Approximating a wavefunction as an unconstrained sum of slater determinants. *Journal of Mathematical Physics*, 49:032107, 2008.
- [34] Stuart J. Berkowitz. On computing the determinant in small parallel time using a small number of processors. *Information Processing Letters*, 18(3):147–150, 1984.
- [35] Marcus Hutter. On representing (anti)symmetric functions. 2020. preprint.
- [36] David Pfau, James S. Spencer, Alexander G. D. G. Matthews, and W. M. C. Foulkes. Ab initio solution of the many-electron schrödinger equation with deep neural networks. *Physical Review Research*, 2:033429, 2020.
- [37] Jeffmin Lin, Gil Goldshlager, and Lin Lin. Explicitly antisymmetrized neural network layers for variational monte carlo simulation. *Journal of Computational Physics*, 474:111765, 2023.
- [38] Aaron Zweig and Joan Bruna. Towards antisymmetric neural ansatz separation. 2022. preprint.
- [39] Stephan Mertens and Cristopher Moore. The complexity of the fermionant, and immanants of constant width. *Theory of Computing*, 9:273–282, 2013.